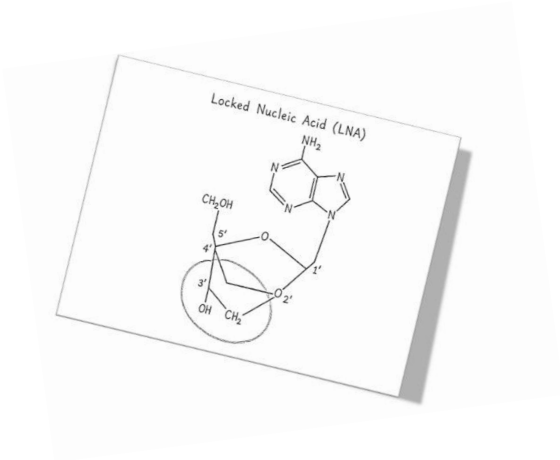
Locked Nucleic Acids (LNA)



Locked nucleic acids were introduced in the late 1990s:

Obika, Satoshi; Nanbu, Daishu; Hari, Yoshiyuki; Morio, Ken-ichiro; In, Yasuko; Ishida, Toshimasa; Imanishi, Takeshi (1997-12-15). "Synthesis of 2'-O,4'-C-methyleneuridine and -cytidine. Novel bicyclic nucleosides having a fixed C3, -endo sugar puckering". Tetrahedron Letters. 38 (50): 8735–8738. Doi:10.1016/S0040-4039(97)10322-7. ISSN 0040-4039.

These nucleosides incorporate a 2′-O,4′-C methylene bridge that locks the ribose ring into a C3′-endo conformation characteristic of RNA duplexes (Koshkin et al., 1998; Obika et al., 1998).

Unlike earlier modifications, LNA technology was a shift toward structure-driven nucleic acid engineering. Constrained ethyl (cEt) chemistry is a type of LNA modification that incorporates a 2′-O-ethyl substituent within a constrained bicyclic framework; used primarily by Ionis Pharmaceuticals. Structural analyses show that cEt and LNA adopt highly similar sugar conformations, while cEt positions its substituents closer to the phosphate backbone, contributing to enhanced resistance to enzymatic degradation (Pallan et al., 2012).

	ΔTm per modification (°C)	Relative affinity vs DNA
DNA (unmodified)	Baseline	
Phosphorothioate (PS DNA)	0	0.8–1×
2′-O-methyl (2′-OMe)	+0.5 to +1.0	1.2–1.5×
2′-fluoro (2′-F)	+1.0 to +2.0	1.5–2×
2′-O-methoxyethyl (2′-MOE)	+0.9 to +1.7	1.5–2×
Locked nucleic acid (LNA)	+3.5 to +8.0	3–10×

LNA as a biological tool

Locked nucleic acid (LNA) chemistry has become an important component of modern PCR-based technologies, particularly in applications requiring high specificity, sensitivity, and reproducibility. The enhanced hybridization affinity conferred by LNA modifications enables the design of short oligonucleotide primers and probes with elevated and tunable melting temperatures (Tm), which improves assay performance across a wide range of targets (Braasch and Corey, 2001; Grunweller and Hartmann, 2007).

Incorporation of LNA residues into PCR primers and fluorescent probes:

- Improved binding to difficult or GC-rich templates
- Shorter primer design without loss of affinity
- Enhanced mismatch discrimination
- Enhanced fluorescence signal upon hybridization
- Enable stable hybridization to short targets

Strategic placement of LNA nucleotides often near the 3′-end can significantly improve allele-specific amplification, making LNA-modified primers particularly useful for single nucleotide polymorphism (SNP) detection and genotyping assays (Di Giusto and King, 2004; Mouritzen et al., 2003). The increased thermodynamic stability also reduces nonspecific amplification by favoring correct primer–template pairing. (Morandi et al., 2007; Valoczi et al., 2004; Castoldi et al., 2006).

Miravirsen is a first-in-class LNA-modified antisense oligonucleotide that targets miR-122, a liver microRNA essential for HCV replication, leading to durable, dose-dependent viral suppression with a strong safety profile and no observed resistance. Its success validated LNA chemistry as a viable and safe platform for RNA-targeting therapeutics.

Miravirsen (Santaris/Roche) – anti-miR-122 (HCV)

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Miravirsen is a 15-nucleotide LNA-modified phosphorothioate antisense oligonucleotide designed to bind and sequester miR-122, a liver-specific microRNA that is essential for hepatitis C virus (HCV) replication. miR-122 normally binds to two conserved sites in the 5′ untranslated region of the HCV genome, enhancing viral RNA stability and replication. By forming a highly stable heteroduplex with miR-122, miravirsen blocks this interaction, functionally inhibiting the microRNA, destabilizing HCV RNA, and suppressing viral replication. In a Phase 2a clinical study, miravirsen was administered as five weekly subcutaneous injections at doses of 3, 5, or 7 mg/kg and demonstrated dose-dependent reductions in HCV RNA, with a 1.2 log10 IU/mL reduction at 3 mg/kg and approximately 2.9–3.0 log10 IU/mL reductions at the 5–7 mg/kg doses. Viral suppression persisted beyond the dosing period, and undetectable HCV RNA was observed in multiple patients receiving higher doses. Key differentiators of miravirsen include the absence of observed viral resistance or escape mutations, activity across HCV genotypes due to conserved miR-122 binding sites, favorable pharmacokinetics with a prolonged tissue half-life, and a lack of cytochrome P450 interactions, which reduces the risk of drug–drug interactions. The safety profile was also favorable, with no dose-limiting toxicities, mostly mild to moderate adverse events such as headache and fatigue, no significant hepatotoxicity or clinically relevant laboratory abnormalities, and reductions in cholesterol levels consistent with the known biology of miR-122.

Cobomarsen is an LNA-based anti-miR-155 therapy for cutaneous T-cell lymphoma that showed encouraging early clinical results, including a favorable safety profile and disease improvement in most patients, supporting its potential as a novel microRNA-targeted cancer treatment.

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Phase 1 clinical trial evaluated Cobomarsen in patients with mycosis fungoides (MF), the most common subtype of CTCL, assessed safety, tolerability, pharmacokinetics, and preliminary efficacy across multiple routes of administration, including intralesional, subcutaneous (SC), and intravenous (IV) delivery at doses up to 900 mg.

Cobomarsen demonstrated a favorable safety profile, with no serious adverse events attributed to the drug. The most reported AE was fatigue, neutropenia, injection-site pain, nausea, pruritus, and headache. Dose-limiting toxicities were limited to Grade 3 pruritus and tumor flare, and the maximum tolerated dose was not reached.

91% showed improvement in disease burden as measured by the modified Severity Weighted Assessment Tool mSWAT -- the percentage of Body Surface Area affected by patches, plaques, and tumors.

Importantly, improvements in clinical endpoints were accompanied by enhanced quality of life and meaningful patient benefits beyond tumor reduction. These results highlight the potential therapeutic potential of LNA-based anti-miR strategies in oncology. The strategy is not without controversy, however. Targeting microRNA can have unintended consequences as the intended target typically regulates hundreds of other genes. Therefore, it is essential to understand the biological consequences of a potential "on-target" tox effect.

RG-101 is a GalNAc-conjugated anti-miR-122 therapy

RG-101 is a GalNAc-conjugated anti-miR-122 therapy that showed strong antiviral activity and good initial tolerability but later revealed safety concerns due to transient hyperbilirubinemia from off-target effects. This led to the development of RG6650, an optimized follow-up compound designed to maintain efficacy while improving safety by reducing interference with bilirubin transport.

RG-101	LNA/MOE/DNA	Clinically strong antiviral, but safety issue
RG6650	cEt/DNA	selected for improved safety

In early clinical studies (phase 1B study), RG-101 was well tolerated after single-dose administration, with no serious adverse events and mostly mild, self-limiting side effects such as fatigue and injection site reactions (van der Ree et al., 2017). Transient increases in alkaline phosphatase and a single case of cholestasis were observed, but there was no consistent evidence of hepatocellular injury or significant bilirubin elevation across the cohort (van der Ree et al., 2017). Later studies of RG-101 demonstrated strong antiviral efficacy but showed transient hyperbilirubinemia that was attributed to off-target inhibition of the MRP2 transporter (van der Ree et al., 2017; Drygin, 2022). Following this finding, Regulus conducted a focused optimization effort to identify next-generation compounds that retained potent miR-122 inhibition while minimizing interference with bilirubin transport (Drygin, 2022). This work involved a screening and structure–activity relationship (SAR) campaign that ultimately leading to the identification of RG6650 (Drygin, 2022).

	RG-101	RG6650 / RG7443
Base	A m5C A m5C m5C A m5U T G U C A C A C T C C A	U C A C A C T C C
Backbone	Full phosphorothioate	Full phosphorothioate
Sugar	E E E E E E E d d L L d L d L d L L d	K K d K d K d K K
Legend	E = 2′-MOE; d = DNA; L = LNA; A = adenine; T = thymine; G = guanine; C = cytosine; U = uracil; m^5C = 5-methylcytosine; m^5U = 5-methyluracil.	K = S-cEt; d = DNA
Conjugation	3′ GalNAc-conjugated	3’ GalNAc-conjugated

EZN-4176

In early clinical development, the LNA gapmer EZN-4176 exemplified the balance between potency and safety that characterizes high-affinity antisense chemistries. In a first-in-human phase I study in patients with castration-resistant prostate cancer, EZN-4176 demonstrated limited clinical activity despite promising preclinical rationale. Treatment was generally tolerated at lower dose levels; however, dose escalation was constrained by hepatotoxicity, with reversible elevations in alanine aminotransferase (ALT) and aspartate aminotransferase (AST) observed as dose-limiting toxicities at 10 mg/kg (Bianchini et al., 2013). These transaminase elevations occurred in the absence of concomitant bilirubin increases, suggesting hepatocellular injury rather than cholestatic dysfunction (Bianchini et al., 2013). Despite achieving systemic exposures consistent with pharmacological activity, no meaningful reduction in androgen receptor (AR) expression or sustained antitumor responses were observed, and the study was discontinued (Bianchini et al., 2013).